

Test-Time Adaptation: Recurrent State vs. Gradient Updates, and Where BDH-CQ Fits

Every ARC-AGI-style task hands a model a handful of input-output demonstration pairs and asks it to apply the same unstated rule to a new input. A frozen, pretrained network cannot do this by pattern-matching alone, because the rule is invented per task. Solving it requires some form of adaptation that happens after pretraining and before or during the answer — this is test-time adaptation, and the field has converged on two structurally different mechanisms for it.

The first mechanism is optimization: treat the demonstrations as a miniature training set and update parameters against them before evaluating the query. The Hierarchical Reasoning Model (HRM) exemplifies this at the architecture level, using two coupled recurrent modules operating at different timescales — a slow module for abstract planning and a fast module for detailed computation — trained end to end on roughly 1,000 examples per task family, reported to outperform much larger chain-of-thought language models on ARC-AGI and on Sudoku and maze benchmarks with only about 27 million parameters. The Tiny Recursive Model (TRM) simplifies this further to a single small weight-tied network that recursively refines a latent state and a candidate answer over a fixed number of steps, reporting 45% accuracy on ARC-AGI-1 and 8% on ARC-AGI-2 with about 7 million parameters — ahead of several much larger language models on the same benchmark. A 2025 follow-up study makes the optimization commitment explicit by comparing full fine-tuning, embeddings-only tuning, and LoRA-style adaptation of a TRM checkpoint: every variant still requires computing a gradient against the specific evaluation task before that task can be answered.

The second mechanism keeps parameters frozen entirely and instead lets a bounded piece of recurrent state absorb the demonstrations as it reads them, with no backward pass at inference. This is the route Pathway's BDH-CQ takes. Built on the Dragon Hatchling (BDH) architecture — a scale-free, brain-inspired network in which attention is reformulated as a Hebbian, synapse-level update rather than a dense pairwise comparison over all tokens — BDH-CQ combines in-context learning with recurrent latent reasoning: demonstrations continuously update the model's recurrent memory, and the model then iterates in a high-dimensional latent space to produce an answer, without generating a verbalized chain of thought. A 150-million-parameter configuration reaches 29.5% pass@2 on the public ARC-AGI-1 evaluation set at a computed inference cost of about \$0.0007 per task. Pathway frames this explicitly as a new point on the cost-accuracy Pareto frontier for that benchmark, not as a new accuracy record — systems that spend more compute per task, such as GPT-5.6-class models scoring around 34%, still score higher in absolute terms, at roughly eleven times the cost. A reported black-box audit by outside collaborators is said to have reproduced the 29.5% figure, which supports the reliability of the number without constituting a full independent replication of the underlying method, since core architectural and training details were not disclosed alongside it.

The mechanism-level distinction is the substantive claim, and it is falsifiable at small scale: a linear associative (Hebbian) memory, built from nothing but repeated outer-product additions, can recover a taught input-output mapping perfectly as long as the number of distinct patterns stays within the dimensionality of its state, and it degrades through measurable cross-talk — not through any change in "how much it has learned" — once that capacity is exceeded. This is the same failure mode that limits fast-weight and linear-attention memories generally, and BDH-CQ's own paper frames its contextual memory in exactly these terms, including the special case where state accumulates additively per demonstration.

Both routes have real, disclosed limitations. HRM's originally reported advantage has since been partly complicated by mechanistic-analysis work showing it can fail on trivially simple puzzles under certain conditions, tied to a violation of the fixed-point assumption the architecture depends on. BDH-CQ's headline number is a cost-efficiency result on one benchmark (ARC-AGI-1), evaluated by its own developer, with the harder ARC-AGI-2 and general mathematical reasoning explicitly named as future work rather than demonstrated capability. Neither figure should be read as evidence that either family of methods generalizes beyond the benchmark it was measured on, and the strongest gains reported for the optimization route (TRM, HRM) required per-task or per-domain hyperparameter tuning that the recurrent-state route does not.

Continue learning: Kosowski et al., "The Dragon Hatchling" (arXiv:2509.26507); Pathway Research, "BDH-CQ: In-Context Learning with Recurrent Latent Reasoning" (arXiv:2608.09888); Wang et al., "Hierarchical Reasoning Model" (arXiv:2506.21734); Jolicoeur-Martineau, "Less is More: Recursive Reasoning with Tiny Networks" (arXiv:2510.04871).